

Daily Research Digest — 2026-04-22

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter.

Synthesis

Executive Overview

Recent preprints highlight advancements in the study of galaxy clusters, galaxies, gravitational lensing, and dark matter through a variety of methods. Key themes include precision measurements using spectroscopic data, detailed analysis of gravitational effects like the kinetic Sunyaev-Zel'dovich effect (kSZ), and novel probes for constraining dark matter models during early epochs. The papers also underscore the importance of high-resolution imaging and modeling in elucidating galactic properties and the role of feedback mechanisms.

Galaxy Clusters

No recent preprints are tagged specifically with galaxy clusters, so this section is sparse. Focus remains on indirect contributions through related topics like dark matter and gravitational lensing.

Galaxies

Recent submissions emphasize detailed studies of galaxies using high-resolution imaging and spectroscopic data from telescopes such as Hubble and ALMA. Papers include investigations into the luminosity functions and spectral energy distributions (SEDs) of active galactic nuclei (AGN), protoclusters, supernova gravitational waves, and accretion processes in young stellar objects (YSOs). These studies aim to disentangle AGN properties from host galaxy emission through morphological decomposition techniques and resolve complex astrophysical phenomena.

Gravitational Lensing

Submissions on gravitational lensing include the analysis of a rare radio changing-look Narrow-line Seyfert 1 (NLS1) galaxy, SDSSJ110546.07+145202.4, which presents unique challenges and opportunities for understanding outburst mechanisms in AGNs. This work provides comprehensive observational data across multiple wavelengths, including the first high-frequency radio observations and X-ray monitoring.

Dark Matter

Recent dark matter papers explore constraints on TeV-scale subcomponents of dark matter through cosmic microwave background (CMB) data and future 21-cm signal detections. Another paper investigates the kinetic Sunyaev-Zel'dovich effect for probing feedback mechanisms in galaxies, offering a novel approach to constrain dark matter models by examining decays into different particle types during early cosmological epochs.

Characterizing Supernova Gravitational Waves

A recent submission explores the potential of supernovae as gravitational wave sources using protoneutron star asteroseismology. The study aims to establish universal relations between gravitational wave signals and physical properties independent of model parameters, which could aid in future detections and interpretations of these events.

Spectral Modeling of YSO Eruptions

Research on young stellar objects (YSOs) focuses on identifying and characterizing FUor outbursts using infrared time-domain surveys and high-resolution spectroscopy. The paper employs Gemini South/IGRINS for follow-up observations, constructing SEDs to fit disk models with both actively accreting inner disks and passively irradiated outer disks.

Cosmological Constraints on Decaying Dark Matter

A recent study revisits constraints from the CMB on decaying dark matter subcomponents composed of TeV-scale particles. The paper re-evaluates these constraints using current datasets and estimates the sensitivity of future 21-cm signal detections, indicating that this method may be more sensitive to certain decay scenarios than previous analyses.

Conclusion

These recent submissions collectively advance our understanding of galactic properties, dark matter models, and gravitational phenomena through a combination of observational data, high-resolution imaging techniques, and novel theoretical approaches. The detailed investigations into the kinetic Sunyaev-Zel'dovich effect and decaying dark matter subcomponents provide promising avenues for future research in cosmology.

Papers to Read

- **"Understanding supernova gravitational waves with protoneutron star asteroseismology"**: Provides insights into using supernovae as gravitational wave sources.
- **"Cosmological constraints on TeV-scale dark matter subcomponents decaying between recombination and reionisation"**: Explores novel approaches to constraining early universe physics through CMB data and future 21-cm signal detections.

These papers offer significant contributions to their respective fields, providing valuable insights into the complex astrophysical phenomena under investigation.

Paper Summaries and Figures

1. Precision Kinematic Sunyaev–Zel’dovich Measurements Across Halo Mass and Redshift with DESI DR2 and ACT DR6: Part II. Bright Galaxy Survey and Emission-Line Galaxies

Focus: galaxies, gravitational lensing

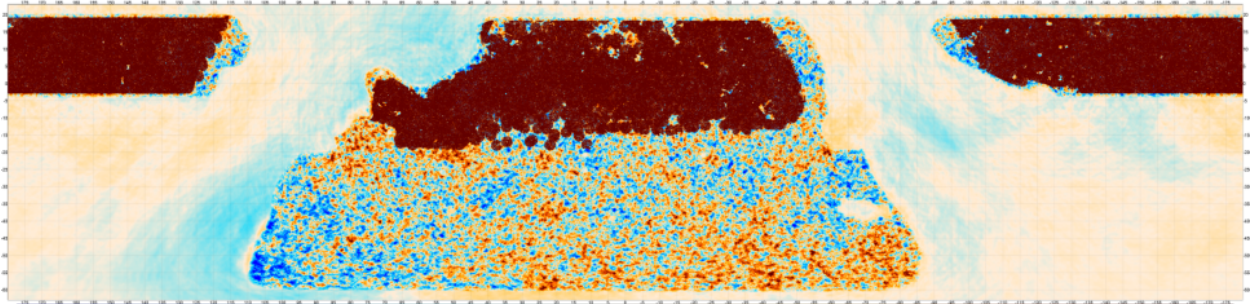
Authors: B. Hadzhiyska, S. Ferraro, F. J. Qu, B. Ried Guachalla, E. Schaan, J. Aguilar, S. Ahlen, D. Bianchi, D. Brooks, F. J. Castander, E. Chaussidon, T. Claybaugh, A. de la Macorra, Arjun Dey, Biprateep Dey, P. Doel, J. E. Forero-Romero, E. Gaztañaga, S. Gontcho A Gontcho, G. Gutierrez, J. Guy, K. Honscheid, C. Howlett, D. Huterer, M. Ishak, R. Joyce, R. Kehoe, T. Kisner, A. Kremin, O. Lahav, M. Landriau, L. Le Guillou, A. Leauthaud, M. Manera, P. Martini, A. Meisner, R. Miquel, S. Nadathur, N. Palanque-Delabrouille, W. J. Percival, F. Prada, I. Pérez-Ràfols, G. Rossi, L. Samushia, E. Sanchez, E. F. Schlafly, D. Schlegel, J. Silber, D. Sprayberry, G. Tarlé, B. A. Weaver, R. Zhou, H. Zou

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PDF: <https://arxiv.org/pdf/2604.19745v1>

Summary: We present the first high-significance spectroscopic stacked kinetic Sunyaev-Zel’dovich (kSZ) measurements of circumgalactic gas profiles for both Bright Galaxy Survey (BGS) and Emission Line Galaxy (ELG) tracers, combining DESI Data Release 2 with ACT Data Release 6. Using reconstructed line-of-sight velocities from the DESI galaxies and high-resolution ACT temperature maps, we detect the kSZ signal at high significance, reaching signal-to-noise ratios of up to ~ 9 for BGS and ~ 7.5 for ELGs in optimal stellar-mass selections. Together with the LRG measurements presented in Paper I, these constitute the most significant kSZ detections from any spectroscopic survey to date. We perform the analysis in both real and harmonic space, obtaining consistent results. By splitting both tracers into stellar-mass bins, we study the scaling of the kSZ amplitude with galaxy properties. Combining the kSZ measurements with ACT Data Release 6 (DR6) CMB lensing maps enables a joint calibration of the galaxy-halo connection and the gas fractions of host halos. For the BGS galaxies, we observe low gas fractions around the virial radius relative to standard expectations, likely attributable to active galactic nuclei (AGN) activity. We find some evidence for higher-mass halos retaining a larger fraction of their baryons, consistent with more efficient feedback in lower-mass systems. For the ELG sample, dominated by blue, star-forming galaxies, we provide the first detection of the gas distribution in ELG host halos. The ELGs appear to exhibit relatively high gas fractions, which points to the possibility of weaker feedback (due to e.g. low AGN and supernova feedback activity) at their mass scale. Finally, we present generalized Navarro-Frenk-White (GNFW) fits to the harmonic-space measurements, providing a compact parametrization of gas profiles for forward modeling in large-scale structure analyses.

Selected figures



2. Precision Kinematic Sunyaev–Zel’dovich Measurements Across Halo Mass and Redshift with DESI DR2 and ACT DR6: Part I. Luminous Red Galaxies

Focus: galaxies, dark matter

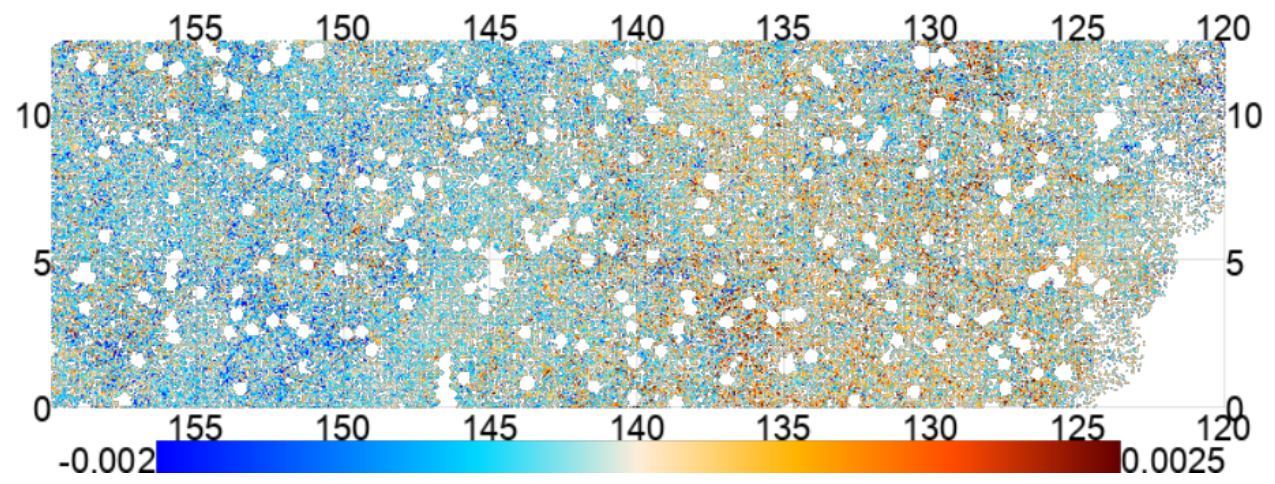
Authors: F. J. Qu, B. Ried Guachalla, E. Schaan, B. Hadzhiyska, S. Ferraro, J. Aguilar, S. Ahlen, A. Baleato Lizancos, D. Bianchi, D. Brooks, R. Canning, F. J. Castander, E. Chaussidon, T. Claybaugh, A. Cuceu, A. de la Macorra, B. Dey, P. Doel, A. Font-Ribera, J. E. Forero-Romero, E. Gaztañaga, S. Gontcho A Gontcho, G. Gutierrez, H. K. Herrera-Alcantar, K. Honscheid, C. Howlett, D. Huterer, M. Ishak, R. Kehoe, T. Kisner, A. Kremin, O. Lahav, M. Landriau, L. Le Guillou, M. E. Levi, M. Manera, A. Meisner, R. Miquel, S. Nadathur, J. A. Newman, W. J. Percival, I. P’erez-R’afols, G. Rossi, L. Samushia, E. Sanchez, E. F. Schlafly, D. Schlegel, M. Schubnell, H. Seo, J. Silber, D. Sprayberry, G. Tarl’e, B. A. Weaver, R. Zhou

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PDF: <https://arxiv.org/pdf/2604.19744v1>

Summary: We present the most precise measurements of the kinetic Sunyaev-Zel’dovich (kSZ) effect around luminous red galaxies to date, detecting the signal at 18σ significance in both harmonic and configuration space. Our analysis cross-correlates 2.4 million spectroscopic LRGs from the Dark Energy Spectroscopic Instrument (DESI) DR2 sample with Data Release 6 (DR6) of the Atacama Cosmology Telescope (ACT). We develop a novel harmonic-space cross-correlation approach using momentum-weighted kSZ templates, yielding nearly uncorrelated bandpowers within a framework consistent with other large-scale structure analyses. By incorporating the LRG halo occupation distribution (HOD) and its uncertainty, we convert measured galaxy gas profiles into halo gas profiles and provide generalized Navarro-Frenk-White (GNFW) fitting profiles, providing empirical targets for tuning feedback efficiency in hydrodynamical simulations and for baryonic modeling in large-scale structure analyses. We find strong evidence that gas profiles do not trace dark matter, providing direct evidence for gas redistribution beyond gravitational collapse. Comparing to hydrodynamical simulations, our measurements favor feedback efficiencies exceeding those in the Battaglia profile, suggesting more efficient gas ejection in group-scale halos than previously predicted. Splitting by redshift, we detect the kSZ signal at $\text{SNR} \approx 5\text{--}10$ in each of four bins and find amplitude evolution consistent with the expected decline in mean halo mass at fixed comoving number density. Splitting by stellar mass, we study the scaling of kSZ amplitude with galaxy properties. Together with BGS and ELG measurements in Paper II, these results span $0.1 \lesssim z \lesssim 1.6$ across three galaxy populations, demonstrating the potential of spectroscopic kSZ to map circumgalactic gas and constrain baryonic feedback.

Selected figures



3. Breaking the UV Luminosity Function Degeneracy: Self-Interacting Dark Matter Constraints from Reionization Topology

Focus: galaxies, dark matter

Authors: Zihan Wang, Huanyuan Shan

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PDF: <https://arxiv.org/pdf/2604.19726v1>

Summary: Self-interacting dark matter (SIDM) is the leading framework resolving small-scale cold dark matter (CDM) crises, yet high-redshift SIDM constraints are fundamentally limited by degeneracies between dark matter microphysics and galaxy formation astrophysics. We demonstrate that the UV luminosity function alone cannot constrain SIDM: star formation suppression from SIDM halo core formation is fully absorbed by modest adjustments to standard astrophysical parameters. We show that 21 cm reionization topology breaks this degeneracy completely, providing a nuisance-immune probe: the SIDM-enhanced duty cycle of ionizing photon escape leaves a morphological signature fully independent of star formation efficiency. Combining JWST UVLF measurements with SKA1-Low forecasts, constant-cross-section SIDM with $\sigma/m \gtrsim 1\text{--}2 \text{ cm}^2/\text{g}$ is either excluded or detectable across all physically motivated star formation coupling strengths. Our results establish a robust new avenue to probe dark matter microphysics in the early Universe.

4. Resolved UV-Optical HST Imaging and Spectral Energy Distribution Modeling of Nearby BAT Active Galactic Nuclei

Focus: galaxies

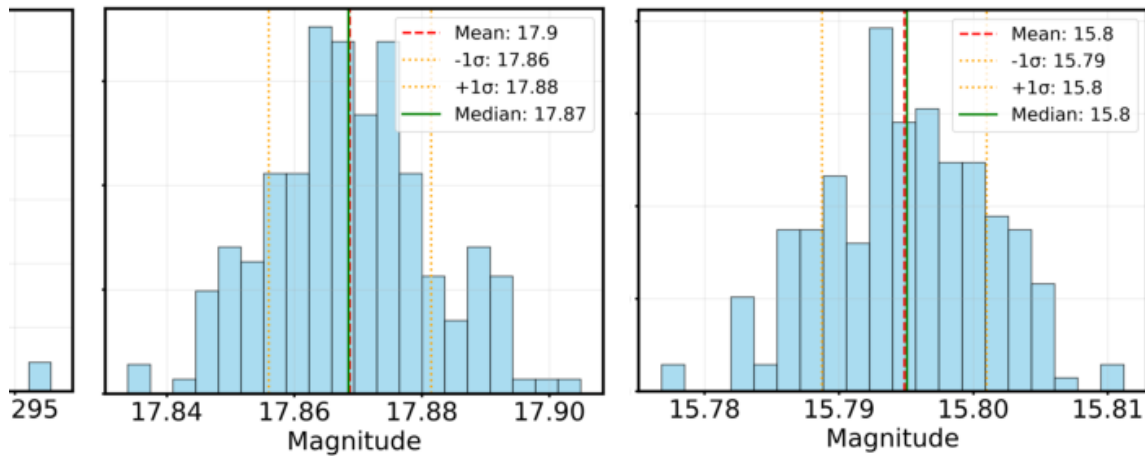
Authors: Connor Auge, Michael Koss, Kriti K. Gupta, Claudio Ricci, Benny Trakhtenbrot, Franz E. Bauer, Ezequiel Treister, Alessandro Peca, Brad Cenko, Kohei Ichikawa, Arghajit Janna, Darshan Kakkad, Richard Mushotzky, Kyuseok Oh, Alejandra Rojas Lilayú, David Sanders, Roberto Serafinelli, Matilde Signorini, Alessia Tortosa, C. Megan Urry

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PDF: <https://arxiv.org/pdf/2604.19674v1>

Summary: We use high-resolution UV-to-optical imaging from the Hubble Space Telescope (HST) to construct spatially resolved spectral energy distributions (SEDs) for seven nearby ($z < 0.07$) hard (14–195 keV) X-ray-selected broad-line active galactic nuclei (AGN) with $L_{\text{bol}} = 10^{43.26} - 10^{45.34} \text{ erg s}^{-1}$. The high spatial resolution of HST, which physically resolves structures on the scale of $\sim 50 \text{ pc}$ at $z = 0.05$, enables the separation of AGN and host-galaxy emission through morphological decomposition with GALFIT, yielding improved measurements of AGN properties compared to those obtained with lower-resolution Swift UV/Optical Telescope (UVOT) data. AGN UV magnitudes derived from HST imaging (e.g., F225W) can differ by more than a magnitude from those from Swift/UVOT UVM2 due to extended nuclear emission. Additionally, the inclusion of high-resolution data at longer wavelengths (e.g., F814W) can significantly affect the resulting SED fit. Comparing fits of accretion disk and extinction models using HST and Swift/UVOT data, we find significant differences in the resulting parameters, with average differences of 2.0 eV in the maximum disk temperature and 2.2 mag in the AGN host-galaxy extinction. These differences ultimately lead to significant changes in bolometric luminosities and X-ray bolometric corrections, with the HST-based fits yielding average increases of $\sim 0.57 \text{ dex}$ and $\sim 0.66 \text{ dex}$ respectively. This demonstrates host-galaxy contamination in unresolved UV-optical data can strongly bias SED-based estimates of disk temperatures, extinction, bolometric luminosities, and X-ray bolometric corrections in AGN. Large-area, high-resolution imaging surveys from Euclid and the Nancy Grace Roman Space Telescope will extend these techniques to much larger AGN samples, enabling uniform, high-precision SED measurements in the near-IR.

Selected figures



5. A Possible Protocluster of Galaxies Serendipitously Discovered in the Field of an Intermediate-Redshift Post-starburst Galaxy

Focus: galaxies

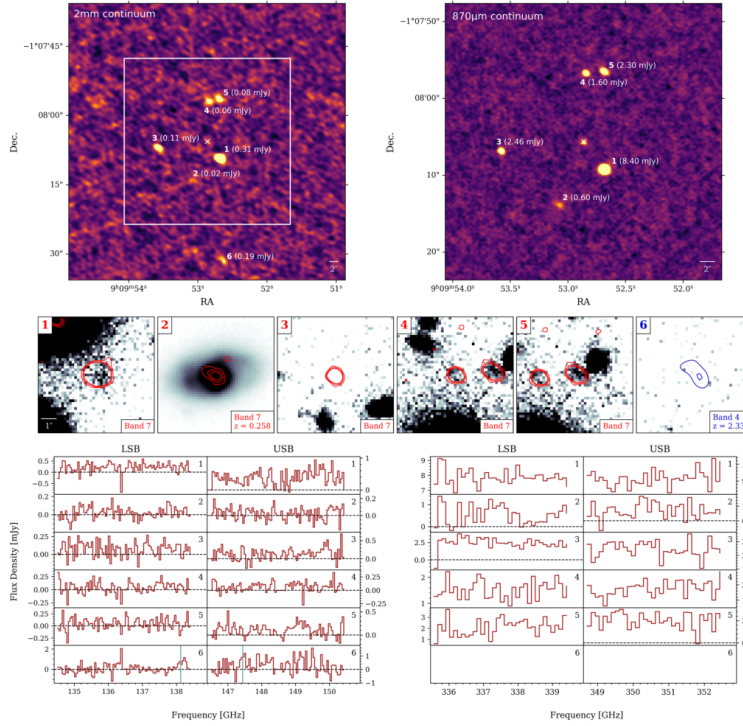
Authors: Mary C. Knowlton, Justin S. Spilker, Rachel Bezanson, Vincenzo R. D’Onofrio, Anika Kumar, David J. Setton, Katherine A. Suess

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PDF: <https://arxiv.org/pdf/2604.19651v1>

Summary: We present the serendipitous discovery of an overdensity of submillimeter galaxies (SMGs) in the field of SDSSJ0909-0108, a massive $z \sim 0.7$ post-starburst galaxy from the SQuIGGLE survey. ALMA observations at 870 μ m and 2mm reveal six galaxies within a 35'' region with flux ratios consistent with emission from dust. Given the rarity of 870 μ m sources and the small field-of-view of ALMA, we speculate that some of these sources are physically associated. None of the sources are at the same redshift as the post-starburst, and four do not have spectroscopic redshifts. We suggest that follow-up optical and/or ALMA observations be carried out to measure redshifts for the galaxies in this potential protocluster environment.

Selected figures



6. MG-NECOLA: A Field-Level Emulator for $f(R)$ Gravity and Massive Neutrino Cosmologies

Focus: galaxies

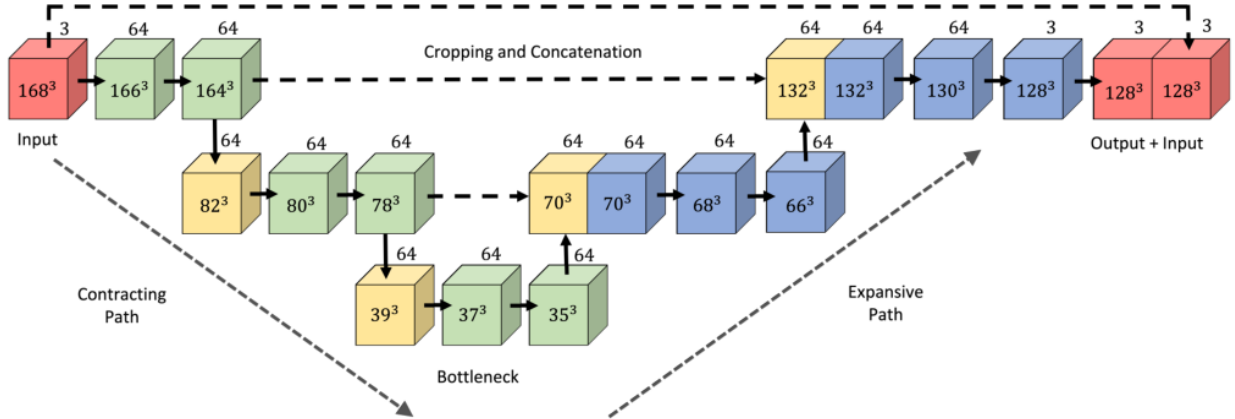
Authors: J. Bayron Orjuela-Quintana, Mauricio Reyes, Elena Giusarma, Marco Baldi, Neerav Kaushal, César A. Valenzuela-Toledo

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PDF: <https://arxiv.org/pdf/2604.19613v1>

Summary: Accurate modeling of non-linear gravitational dynamics is essential for constraining extensions to the standard cosmological model using large-scale structure observations. While high-resolution N -body simulations provide the required fidelity, they are computationally prohibitive for the large ensembles needed to analyze Modified Gravity (MG) scenarios. We present MG-NECOLA, a field-level emulator based on a convolutional neural network that upgrades fast, approximate MG-PICOLA simulations to near- N -body accuracy at a fraction of the computational cost. Trained on a suite of QUIJOTE_MG simulations for $f(R)$ gravity, MG-NECOLA achieves nearly sub-percent accuracy ($\lesssim 1\%$) in both the matter power spectrum and bispectrum up to $k \simeq 1 \, h \text{Mpc}^{-1}$. Crucially, although being trained on a fixed cosmology, the network generalizes robustly to cosmologies outside its training manifold keeping the error below 5%. It successfully recovers the General Relativity limit (CDM) without introducing spurious MG signals and accurately captures the power suppression induced by massive neutrinos ($M \leq 0.4 \text{ eV}$), despite being trained on cosmologies with massless neutrinos. The pipeline delivers a speed-up factor of $\sim 1500\times$ relative to full N -body runs, generating a high-fidelity realization in $O(10^3)$ CPU seconds compared to $O(10^6)$ for the baseline. This accuracy-efficiency trade-off establishes MG-NECOLA as a powerful tool for generating the massive mock catalogs required for next-generation galaxy surveys.

Selected figures



7. Understanding supernova gravitational waves with protoneutron star asteroseismology

Focus: galaxies

Authors: Hajime Sotani

Published: 2026-04-21T15:10:34+00:00

PDF: <https://arxiv.org/pdf/2604.19557v1>

Summary: Supernovae are one of the most promising gravitational wave sources. But, since the system of the supernovae is nearly spherically symmetric, the expected gravitational waves from them are relatively weak, compared to the case of the compact binary mergers. Thus, at least using the current gravitational wave detectors, only the gravitational waves from a supernova that occurred in our galaxy could be detected. To reliably extract information from gravitational waves originating from such a low event rate, thorough preparation is essential. However, because supernova gravitational waves strongly depend on model parameters, such as progenitor mass and the equation of state for dense matter, it may be difficult to extract physical properties even if the gravitational waves are detected. The universal relations between gravitational-wave signals and physical properties, independent of model parameters, are important for solving this difficulty. To discuss such a universal relation, in this article, we systematically examine the protoneutron-star oscillation frequencies with the linear analysis, the so-called asteroseismology, and compare them with the gravitational wave signals in the simulations.

8. Characterizing and spectrally modeling embedded FUor eruptions in the near-infrared

Focus: galaxies

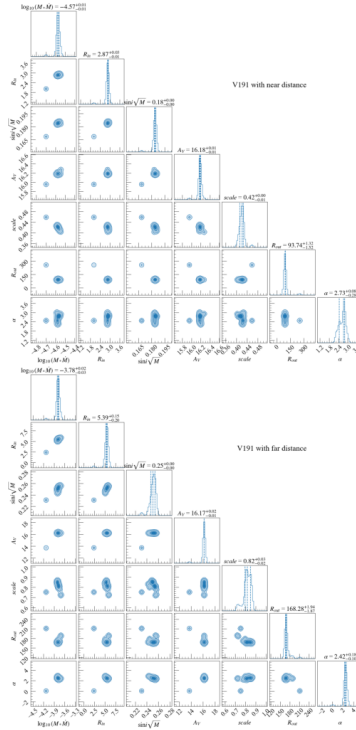
Authors: Jiaxun Li, Tinggui Wang, Zheyu Lin

Published: 2026-04-21T15:07:07+00:00

PDF: <https://arxiv.org/pdf/2604.19551v1>

Summary: Context. Episodic accretion in young stellar objects (YSOs) is thought to play a critical role in addressing the "luminosity problem" associated with star formation. However, optical surveys tend to bias against sources that are heavily obscured. Infrared time-domain surveys, such as unTimely WISE, facilitate the identification of such sources within the dense star formation regions of our Galaxy. Aims. We aim to systematically identify and characterize FUor outbursts in infrared-selected YSOs using high-resolution spectroscopy and detailed disk modeling. Methods. We conducted follow-up high-resolution spectroscopy with Gemini South/IGRINS for four FUor candidates discovered in infrared time-domain surveys. Using a combination of photometric and spectroscopic observations, we constructed spectral energy distributions and fit them with a disk model that incorporates an actively accreting inner disk together with a passively irradiated outer disk. Results. All objects show CO and H₂O absorption bands at 2.3 μ m, and their positions in the Na + Ca versus CO equivalent width diagram further corroborate their classification as FUors. The best-fitting model spectra closely match both the observed spectral features and the overall continuum, providing additional confirmation of the FUor classification. The best-fit models reveal high extinction values ($A_V = 10$ -20 mag), with $\dot{M}_*$ comparable to those of classical FUors such as FU Orionis. Among 18 sources initially selected via infrared light curves, 6–7 out of 8 with available spectra exhibit FUor characteristics, implying a high selection efficiency.

Selected figures



9. SDSSJ110546.07+145202.4: The first long-duration radio changing-look NLS1 galaxy

Focus: gravitational lensing

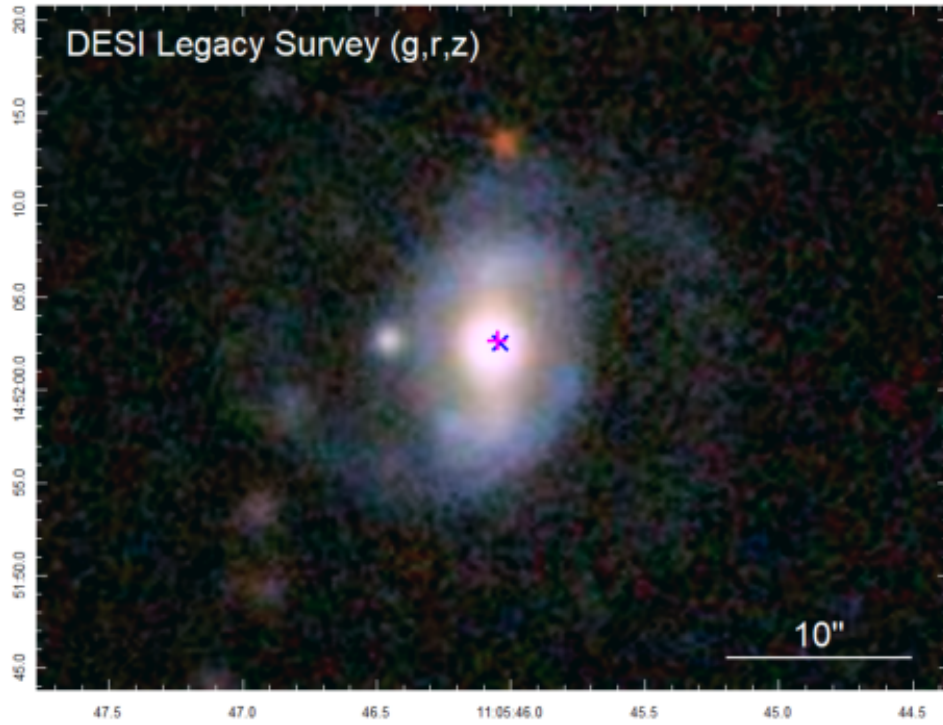
Authors: S. Komossa, D. Grupe, A. Kraus, P. G. Edwards, E. F. Kerrison, K. Rose, R. Soria, T. An, M. J. Hardcastle, K. E. Gabanyi, S. Panda, D. W. Xu, J. Wang, S. Frey, A. Mezosi

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PDF: <https://arxiv.org/pdf/2604.19435v1>

Summary: SDSSJ110546.07+145202.4 stands out as a unique radio changing-look Narrow-line Seyfert 1 (NLS1) galaxy that has brightened dramatically and shows an exceptionally long duration of its "on" phase. We present the first high-frequency radio observations, the first simultaneous radio spectral energy distributions (SEDs), the first optical–UV–X-ray SEDs, and the first X-ray monitoring and spectroscopy of this recently discovered event. Importantly for understanding the nature of the outburst, we show that the X-ray spectrum is soft with a photon index $\Gamma_X=2.5$; line-of-sight absorption and extinction are low or absent; the radio SED is peaked at low frequencies ~ 2 GHz; and the radio outburst emission is very long-lived ($t > 8$ yr) and roughly constant. The softness of the X-ray spectrum, low supermassive black hole (SMBH) mass, and high Eddington ratio all corroborate the optical NLS1 classification. We discuss multiple outburst scenarios, including lensing, absorption, a binary SMBH merger, a long-duration giant-star tidal disruption, a newly ignited active galactic nucleus (AGN), and an accretion-rate change. While most of them can be either excluded or are deemed too rare and lack positive evidence so far, most or all types of these transients are expected to be detected in ongoing VLA and upcoming SKA surveys. SDSSJ110546.07+145202.4 itself is well explained by an accretion rate change that triggered the powerful radio jet emission. The low redshift and SMBH mass of this system offer a unique perspective of the physical processes of radio-jet ignition that are expected to operate in the early Universe around growing SMBHs.

Selected figures



10. Cosmological constraints on TeV-scale dark matter subcomponents decaying between recombination and reionisation

Focus: dark matter

Authors: Markus R. Mosbech, Cristina Benso, Felix Kahlhoefer

Published: 2026-04-21T08:07:34+00:00

PDF: <https://arxiv.org/pdf/2604.19198v1>

Summary: The Dark Ages and the Cosmic Dawn are an untapped well of information about the particle physics properties of dark matter, which may become accessible with future radio telescopes able to probe the 21-cm signal from atomic hydrogen. In this work we study the impact on cosmological observables of a dark matter subcomponent composed of TeV-scale particles that decay into electrons, photons or neutrinos with a lifetime shorter than the age of the universe. We re-evaluate constraints from the Cosmic Microwave Background (CMB) on these scenarios using the most recent data sets and estimate the sensitivity of future detections of the global 21-cm signal. Our main result is that the latter is potentially more sensitive to the effects of decaying dark matter with a lifetime $\tau \gtrsim 10^{15}$ s. This effect is strongest for the case of decays into neutrinos due to the different spectral distribution of the injected electromagnetic energy. For DM masses well above the TeV-scale, these differences become less pronounced and the sensitivity of both the CMB and the 21-cm signal depend primarily on the total amount of injected electromagnetic energy.